

The scope of the policy imposed on the device by the mobile device management solution is, however, limited by the platform hooks provided by the underlying mobile platform for managing the device.

Sadly, the typical security concerns of an enterprise do not get covered in the built-in device management support that the default Android platform provides. However, the open source nature of Android and its easy customisation empowers us to extend the OS to support the specific needs of enterprise usage. Here is an example. In February 2013, American Airlines (AA) deployed 17,000 Samsung Galaxy Notes to its flight attendants for better management of their crew functions, resulting in a better passenger experience. It customised Android to block the phone dialling and camera functions. It removed certain non-essential apps from the platform and pre-bundled various applications that were related to its crew operations. And most important, it chose the Samsung Galaxy Note because it is built on an enterprise ready Android platform, i.e., the Samsung for Enterprise (SAFE) platform. This offers a wide range of device management capabilities, thereby going much beyond the basic device management capabilities of the default Android platform. Rich device management capability, a favourable licence, endless customisation options and cost-effectiveness were the key enablers of this success story.

Given below are some of the enterprise-specific device management requirements that drive the need for an enterprise-ready Android platform.

Communication management: An enterprise systems administrator should be able to whitelist or blacklist outgoing SMS or phone calls for a user. Totally disabling outgoing SMSs or phone calls may also be required at times.

Application management: The sys admin should be able to whitelist or blacklist the applications that can be installed on the user device.

Security management: Data security is the prime concern for any enterprise. The data stored in encrypted form on the device and memory cards should be accessible to only authorised personnel through the proper application channels.

Second, a 'remote wipe' for selected users (enterprise user space) should be possible in case of a device being lost to ensure that the enterprise's data doesn't get into the wrong hands.

Another important function for some enterprises that want to monitor user devices is that all network communication be necessarily through a VPN tunnel. In other words, the user should not be able to carry out any network related communication if the user is not logged on to the VPN client.

Last of all, the sys admin should be able to enable or disable certain accessories and features depending on the degree of security required for the different types of enterprise data. For example, the admin may need to block Internet

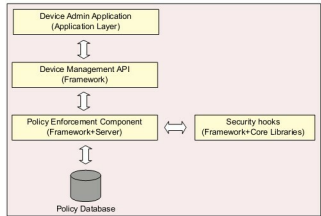


Figure 1: Policy management framework architecture

tethering, Bluetooth, Wi-Fi, the use of an SD card, USB, screen capture, OTA firmware upgrade, the camera, GPS, etc.

Expense management: Enterprises usually have certain budgetary limitations on the data and phone call usage of users. The admin should be able to ensure a role- or user-based data and phone usage quota along with roaming restrictions that control device usage while travelling.

Building an enterprise-ready Android platform

The high level architecture of an enterprise-ready Android platform with an enhanced policy management framework is depicted in Figure 1.

Security hooks

Security hooks are the deterministic control points for a feature in the Android stack. They are uniquely identified in the specific Android version code base and feature specific policy enforcement checks that would be introduced at that point. Every hardware or software feature in a device can be controlled depending on the device management needs of the enterprise. For example, security hooks can be identified for controlling the use of Bluetooth, Wi-Fi, SMS, phone calls, etc, as per policy requirements.

Policy enforcement component

The policy enforcement component provides support for managing and enforcing various policies on the device. It is the core component of the enterprise-ready Android platform and has two major responsibilities.

Policy management: Device admin APIs use this sub-component for reading and updating policy configuration on the device. Policies set on the device are persisted by this subcomponent on it, and are made available for policy enforcement workflow when triggered.

Policy enforcement: The policy enforcement component gets invoked every time a feature that supports some policy is being used. For example, if you are sending an SMS and there is a 'contact whitelisting' policy for outgoing SMS, then the policy enforcement component